

WED - Feb 12 - 2025

Topics: review of gas dynamics, nozzle problem
non-isentropic flow:

normal-shock $\rightarrow$ note that shock thickness is around
200 nm, so it is negligible.

1st law: $h_{o2} - h_{o1} = \dot{q} - \dot{w} \rightarrow h_{o1} = h_{o2} \rightarrow T_{o1} = T_{o2}$

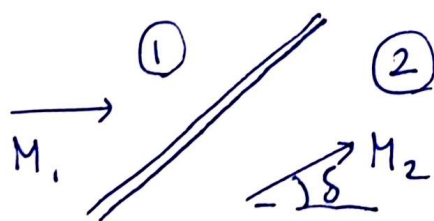
2nd law: $\Delta s = s_{o2} - s_{o1} = c_p \ln\left(\frac{T_{o2}}{T_{o1}}\right) - R \ln\left(\frac{P_{o2}}{P_{o1}}\right)$

$\Rightarrow \Delta s = -R \ln \frac{P_{o2}}{P_{o1}} \Rightarrow \frac{P_{o2}}{P_{o1}} = e^{\frac{-\Delta s}{R}} < 1$

P_o drops across shock. P_o is a measure of useful work, so the work capacity of flow decreases as shock occurs.

① ②	$T_{o1} = T_{o2}$	$P_2 > P_1$	"Normal shock"
$M_1 \rightarrow$	$P_{o2} < P_{o1}$	$u_2 < u_1$	
$M_1 > 1$	$T_2 > T_1$		

N.S



oblique shock

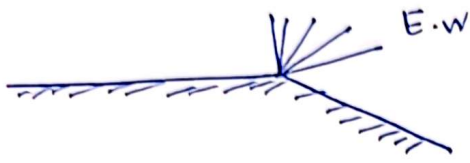
shock angle
 $\sin^{-1}\left(\frac{1}{M}\right) < \delta \leq \frac{\pi}{2}$
 δ : flow angle $\neq 0$

"oblique shock"

(P-1)

If the back pressure is high, flow can not be converted

"Expansion waves" (E.W)

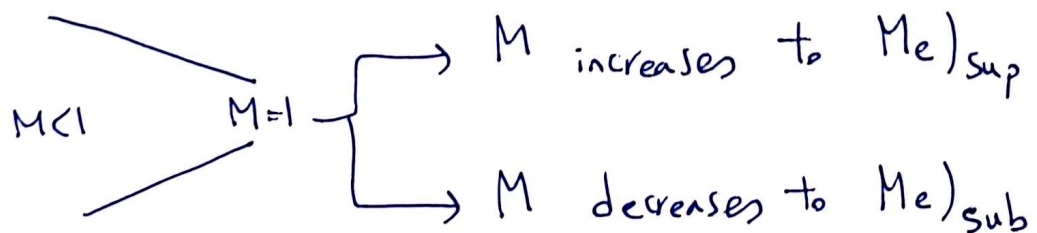
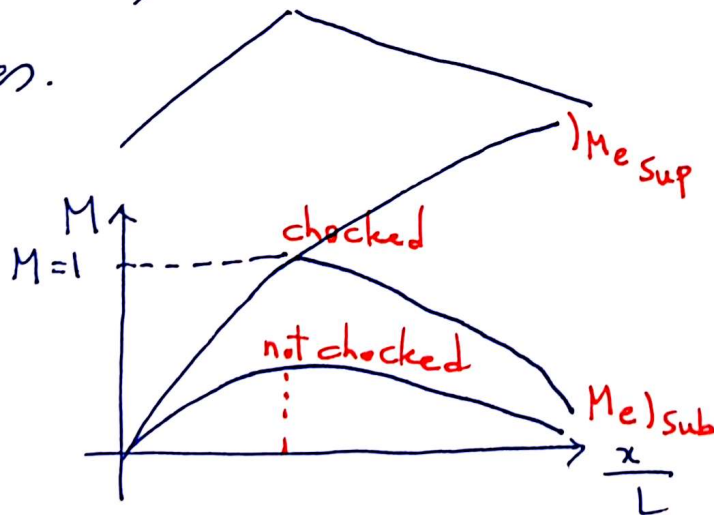


E.W occurs in supersonic flow when area increases or pressure need to be dropped.

Now, let's review flow in "nozzles"



So, in a convergent-divergent nozzle, if the flow does not choke, Mach decreases.



If the back pressure is high, flow can not be converted to supersonic regime.

① $M_e = f\left(\frac{A_e}{A^*}\right) \rightarrow$ read subsonic M_e at $\frac{A_e}{A^*}$, put this M_e in $\frac{P_0}{P_e} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}}$, get $\left(\frac{P_0}{P_e}\right)_{sub}$

(Note for subsonic flow, $(P_e = P_b = P_a)$) $\underbrace{\hspace{10em}}_{P_e = P_b = P_a}$

② read supersonic M_e from $\frac{A_e}{A^*}$, put in $\frac{P_0}{P_e} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}}$

get $\left(\frac{P_0}{P_e}\right)_{sup}$. Note that $\left(\frac{P_0}{P_e}\right)_{sub} \ll \left(\frac{P_0}{P_e}\right)_{sup}$ as $P_{e,sub} \ll P_{e,sup}$

If back pressure is $\left(\frac{P_0}{P_e}\right)_{sub}$ $\rightarrow$ you have $M_{e,sub}$

" " $\rightarrow$ $\left(\frac{P_0}{P_e}\right)_{sup} \rightarrow$ you have $M_{e,sup}$

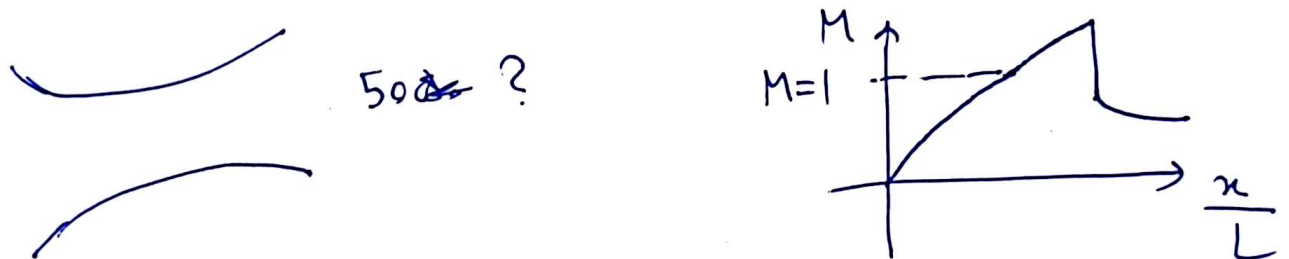
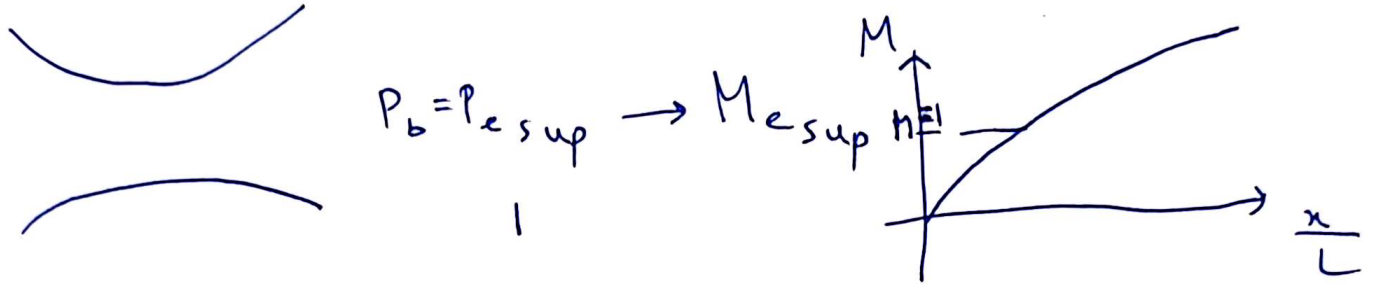
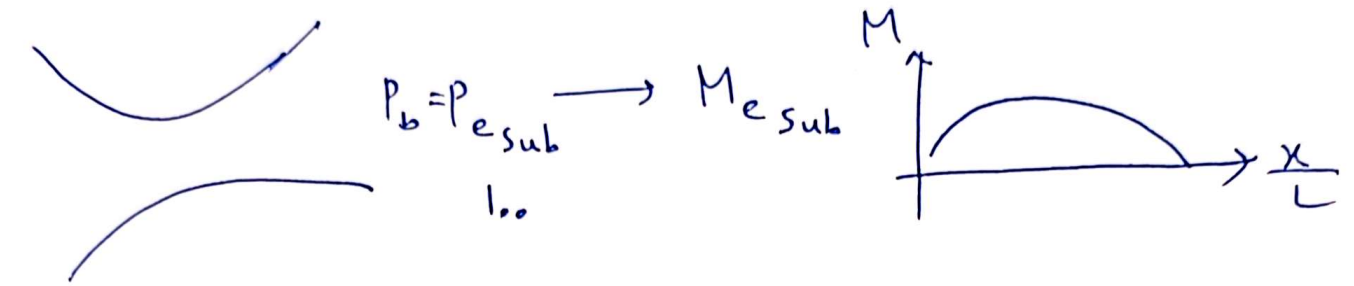
Note $\underbrace{P_1 + \frac{\gamma_1}{2} \rho_1 u_1^2}_{I/A} = \underbrace{P_2 + \frac{\gamma_2}{2} \rho_2 u_2^2}_{I/A}$ (Momentum Eq)

$$P_1 \left(1 + \frac{\gamma_1}{P_1} \rho_1 u_1^2\right) = P_2 \left(1 + \frac{\gamma_2 \rho_2 u_2^2}{P_2}\right) = P_1 (1 + \gamma M_1^2)$$

$\rightarrow$ So momentum Eq $P_1 (1 + \gamma M_1^2) = P_2 (1 + \gamma M_2^2)$ or

$$\boxed{P(1 + \gamma M^2) = cte} \Rightarrow P \uparrow \text{ as } M \downarrow \text{ \& } P \downarrow \text{ as } M \uparrow$$

If $P_{e|sup} < P_b < P_{e|sub}$,



If $P_{e|sup} < P_b < P_{e|sub} \rightarrow$ A normal shock appears in the nozzle.

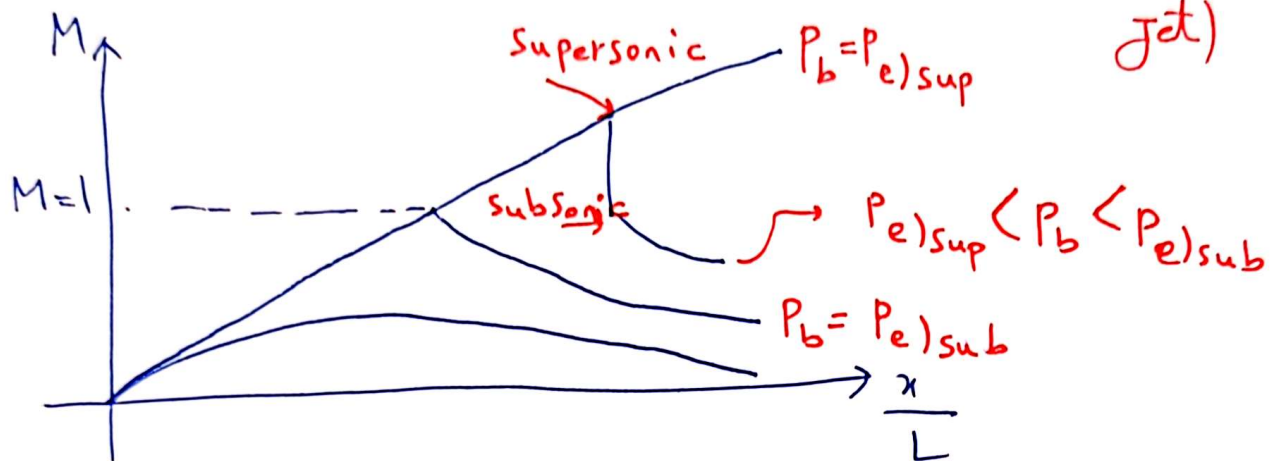
As P_b decreases, the shock approaches the exit. At one specific P_b , i.e., $P_{e|shock@outlet}$, shock occurs at outlet.

If $P_b < P_{e|shock@outlet} \rightarrow$ normal shock converts to oblique shock (over-expanded flow)

If $P_b = P_{e|sup} \rightarrow$ (perfectly expanded flow)

(P = 1)

If $P_b < (P_e)_{sup} \rightarrow$ Expansion waves form outside of the nozzle (under-expanded jet)



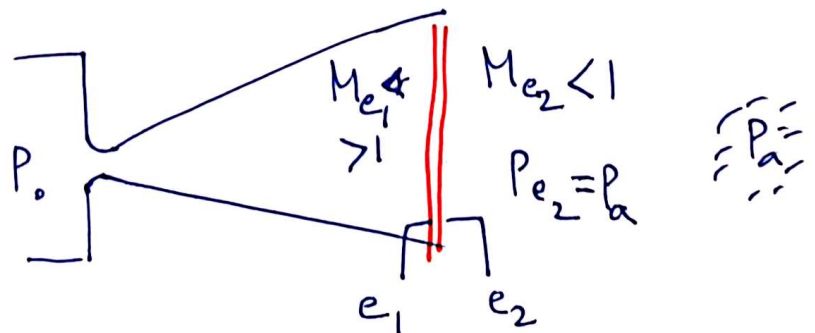
Note: the same $\frac{A_e}{A^*}$ is used in the above fig., but we have different M_e

For rocket nozzles, $P_b = P_a$, thus $\frac{P_0}{P_a}$ & $\frac{A_e}{A^*}$ governs the M_e .

Note: $M_e < 1 \rightarrow P_e = P_a$

Note: $\frac{P_e}{P_e} < \frac{P_e}{P_a} < \frac{P_e}{P_a} \rightarrow$ shock in nozzle

If $\frac{P_0}{P_a}$ further increases, shock moves to the exit plane of the nozzle



(P-51)